

Yubo Du

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Google Scholar: <https://scholar.google.com/citations?user=q8kKxSYAAAAJ&hl=en>

RESEARCH INTERESTS

Compilers, system performance optimization, memory safety, systems security, AI infrastructure

EDUCATION

University of Pittsburgh

Ph.D., Electrical and Computer Engineering

2022/01 – present

Advisor: Prof. [Jun Yang](#)

Vanderbilt University

M.S., Computer Science

2020/01 - 2021/12

Advisor: Prof. [Gautam Biswas](#)

University of Science and Technology of China

B.S., Physics

2014/08 - 2018/06

WORK EXPERIENCES

Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences 2018/08 - 2019/04

Research Assistant

Reconstructing 3D Information from 2D Images with Deep Learning Models

RESEARCH EXPERIENCES

- University of Pittsburgh Advisor: [Dr. Jun Yang](#) 2023/01 - present
Defending Against Temporal Memory Safety Vulnerabilities
 - Developed performance- and memory-efficient techniques for use-after-free (UAF) detection and prevention in C/C++, while preserving existing security guarantees.
 - Proposed *reuse-time tracking (RTT)*, reducing the performance overhead by up to **10×** through eliminating explicit metadata propagation.
 - Designed *fast pointer nullification (FPN)*, achieving up to **19×** lower memory overhead by invalidating dangling pointers at a coarse-grained level.
 - Implemented compiler instrumentations using LLVM and runtime detection/prevention libraries in C/C++.
 - Published work in **ICS 2024** and **NDSS 2026**.
- Vanderbilt University Advisor: [Dr. Gautam Biswas](#) 2021/03 - 2021/12
Differential Neural Architecture Search for Recurrent Neural Network
 - Expanded the search space of RNN to relax the constraints in RNN architecture
 - Proposed a novel framework that increases the network topology progressively
 - 1% - 17% less relative square error than SOTA on time series prediction tasks
 - 1.5X higher training efficiency than Differentiable Architecture Search (DARTs)

PROJECT EXPERIENCES

- University of Pittsburgh Advisor: [Dr. Peipei Zhou](#) 2022/01 - 2022/06
Multimodal Real-time Sentiment Analysis on Embedded Devices
 - Designed a multimodal transformer pipeline for real-time sentiment analysis combining video and audio modalities.
 - Reduced inference latency and memory footprint through model and pipeline tuning for deployment on Jetson TX2.
 - [3rd Place, DAC University Demonstration Competition, Design Automation Conference \(DAC\), 2022.](#)
- University of Pittsburgh Advisor: [Dr. Peipei Zhou](#) 2022/07 - 2022/12
Deep learning for EEG Signal Processing
 - Transformer-based deep learning architecture
 - Heuristic-based differential signal preprocessing
 - 4%-10% higher accuracy than SOTA deep learning methods for EEG signal

RESEARCH PAPERS

- [NDSS'26] **Yubo Du**, Youtao Zhang, Jun Yang, "Fast Pointer Nullification for Use-After-Free Prevention," Proceedings of the 34th Network and Distributed System Security Symposium, NDSS 2026, February 23 - 27, San Diego, CA, USA.
- [ICS'24] **Yubo Du**, Yanan Guo, Youtao Zhang, Jun Yang, "RTT-UAF: Reuse Time Tracking for Use-After-Free Detection," Proceedings of the 38th ACM International Conference on Supercomputing, ICS 2024, June 4 - 7, Kyoto, Japan.
- [FPGA'23] Jinming Zhuang, Jason Lau, Hanchen Ye, Zhuoping Yang, **Yubo Du**, Jack Lo, Kristof Denolf, Stephen Neuendorffer, Alex Jones, Jingtong Hu, Deming Chen, Jason Cong, Peipei Zhou, "CHARM: Composing Heterogeneous Accelerators for Matrix Multiply on Versal ACAP Architecture," Proceedings of the 2023 ACM/SIGDA International Symposium on Field-Programmable Gate Arrays, FPGA 2023, February 12 - 14, Monterey, CA, USA.
- [ITS'20] Yanling Tian, **Yubo Du**, Qieshi Zhang, Jun Cheng, Zhuo Yang, "Depth Estimation for Advancing Intelligent Transport Systems Based on Self-Improving Pyramid Stereo Network," *IET Intelligent Transport Systems*, ITS 2020.

HONORS AND AWARDS

- [DAC Young Fellow, 2022](#)
- [DAC University Demonstration 3rd Place Award, 2022](#)

ACADEMIC SERVICES

- **Artifact Evaluation**

- MLSys, 2022
- ASPLOS, 2026
- **Subreviewer**
 - MLSys, 2022
 - CODES+ISSS, 2022
 - ASP-DAC, 2023

SKILLS

- **Programming Languages**
C/C++, Python, MATLAB
- **Frameworks**
LLVM, PyTorch